# Replacing ATTOM, Smarty and Lob nationwide

A national product decision grounded in Pantopus's web, iOS, Android and backend code, both proposals, nationwide source documentation and checks of public data.

**Recommendation**

Pantopus can replace all three named vendors for a nationwide product. Own the national address index, property model, integrations and household proof. Secure rights to nationwide postal and property inputs, and qualify physical fulfillment before ending the existing services.

## Which proposal wins?

Choose the infrastructure PDF as the base for a nationwide build. It addresses a country-scale registry, property types, national bulk sources and measurable coverage. Add the Word proposal's printer, postage, contract-exit and operating-cost analysis. Both proposals start locally and both miss dependencies in the running app.

| Dependenc | Recommended exit | Experience to preserve or deliver |
|---|---|---|
| **ATTOM** | Acquire nationwide parcel and structural coverage. Improve individual fields from cleared public sources. Source or explicitly defer advanced fields. | Editable Add Home details and source-dated property facts. Owner estimates, taxes and exemptions keep distinct meanings. |
| **Smarty** | Replace the postal evidence behind the existing validation routes. Keep a licensed check for critical decisions. | A missing apartment prompts a correction. A service outage offers recovery. Neither can grant another household's access. |
| **Lob** | Exit after a second fulfillment route handles both verification and neighbor invitations, including in-flight pieces. | A real card with the correct unit, honest status, safe retry, code redemption and recipient opt-out. |

## First investment

Prove the national stack across all 50 states and D.C., with separate coverage rows for each inhabited U.S. territory and military mailing formats. Use representative markets for validation. Every geography gets an address-entry and recovery path from the start. Publish coverage by county and field before promising a result.

Repository snapshot, 1c5ddb658c · National parcel/field coverage · USPS delivery point validation · ATTOM evaluation terms

# Where each proposal earns its place

The Word document tackles the business decision. The PDF gives the engineering team a way to prove a replacement. Their most important disagreements can be resolved against the app.

| Question | Word proposal | Infrastructure PDF | Decision for Pantopus |
|---|---|---|---|
| **All three subscriptions** | Includes print/postage, provider exit, protected addresses and economics. | Page 4 explicitly scopes Lob to address APIs and excludes print and mail. | Word wins on physical-mail scope. Both verification flows and marketing invitations need a fulfillment plan before Lob exits. |
| **App fit** | Models Lob checks and Smarty autocomplete/rooftop as the baseline bill. | Models capabilities broadly and proposes new APIs and entities. | Rebuild from actual routes. Mapbox handles shared typeahead. Google validates and geocodes. Smarty supplies postal and optional unit evidence. |
| **Data and identity** | Independent source families, clean rebuild and separate household evidence. | Stronger unit/building/parcel distinctions, temporal facts, partial results and ID lineage. | Use PDF semantics and Word provenance controls. Reuse existing HomeAddress IDs and private claims. |
| **Evidence and cost** | Coverage ranges, MAU ratios, vendor prices and payback depend on assumptions. | Useful pilot protocol and statistical bounds. Cost, staffing and launch timings are planning ranges. | Measure Pantopus traffic, bills, county field coverage, user outcomes and support labor before setting a payback date. |
| **Scope and risk** | Phone matching, passive location and external API sales add cost or risk. | Models national search and bulk data, but postpones the full U.S. rollout past a local beta. | Defer identity profiling and API resale. Prioritize renters, owners and invite recipients across representative U.S. geographies. |

Use Replacing Lob, Smarty and ATTOM for mail, contracts and economics (Word, pp. 8–10 and 16–20). Use Pantopus Address & Property Infrastructure for property types, postal licensing, source acquisition and evaluation (PDF, pp. 4–13 and 19–25). Replace both local-first schedules with national acquisition and acceptance gates.

Smarty agreement · USPS local engine · Shared geographic provider · Actual validation pipeline

# Smarty sits on the Add Home path

Removing the Smarty key can stop the validation of a new Home. Replacing the search box alone would leave that failure in place.

| App step | What the current code does | What the replacement must do |
|---|---|---|
| **Find an address** | Shared /api/geo/autocomplete and /resolve use the Mapbox provider. /homes/check-address separately looks up existing Homes and canonical hashes. | Keep the clients' suggestion IDs, coordinate format, signed-out budgets and conflict privacy. The nationwide search index can replace typeahead separately. |
| **Validate or add a unit** | Google normalizes components and geocodes with USPS CASS requested. Smarty US Street returns DPV, residential/commercial indicator, missing secondary, commercial mailbox, vacancy and footnotes. | Preserve the existing verdict and next action on web, iOS and Android. Distinguish a service failure, missing unit, unknown unit, PO box, business, mixed use and conflicting household. |
| **Optional enrichment** | Smarty secondary count/list can check a submitted unit. Its rollout switch defaults off. Google Places and ATTOM parcel classification have separate switches. | Use an authorized unit inventory where available. A count cannot verify a named unit. Keep each switch independent and measure exact-unit false acceptance. |
| **Create, claim and check mailbox** | Create revalidates or uses a qualifying recent canonical match in the outage path. A high-score OK can mark an AddressClaim verified. Mailbox Check reads stored postal flags and the caller's own postcard. | Check age and purpose on every reuse. Keep address-quality results separate from occupancy authorization. Preserve the manual/no-unit path and the permission checks. |

Reproduced locally: for synthetic residential input, Smarty DPV D with footnote N1 went through the actual parser and classifier as OK (0.85) and offered a mail code. N1 means a required unit was omitted. A blank DPV also returned OK (0.75). The claim route's OK threshold is 0.8. No live vendor was queried or household privilege exercised. Both cases belong in the exit gate.

The shortest Smarty exit experiment uses the Google adapter already in Pantopus: it requests USPS CASS and parses DPV, CMRA and vacancy. Qualify its coverage and use rights, and supply or explicitly mark unknown the RDI and secondary signals it does not currently map. The existing route still insists on both credentials.

Postal adapter · Decision engine · Claim threshold · Home creation · Secondary lookup · Google USPS fields · DPV and N1 definitions

# ATTOM reaches well beyond Add Home

The backend calls three ATTOM products: property detail, an automated valuation model (AVM), and ZIP sales trends. A replacement needs an explicit choice for each downstream fact.

| Current consumer | Fields or behavior | Replacement policy |
|---|---|---|
| **Add Home and parcel checks** | Suggested type, rooms, floor/lot area, year built. Optional parcel classifier uses land use, parcel ID, building count and residential/nonresidential unit counts. | Prefill independently sourced fields with provenance. Preserve editable household entries. Unknown fields stay empty. Do not infer bedrooms from a model or units from a parcel count. |
| **Home, Place, Today** | Verified-owner Property details has sections for structure, construction, utilities, parking, lot, records, assessment, sale and value. Place uses a role-filtered summary. Today's house illustration can use the property shape. | Map each displayed field, unit, source date and definition. Continue plain house/fallback visuals. Preserve owner-only records and pending-owner/renter/guest distinctions. |
| **Money and neighborhood signals** | Owner value/range/confidence, ZIP trend, tax/exemption status and assessment comparisons. AI property profile, Neighborhood Pulse and generated neighborhood facts also read or label property inputs. | Source AVM separately or show unavailable. Distinguish assessment, last sale and market estimate. An empty exemption field means unknown. Never turn missing trend into "flat." |
| **Owner claim review** | Optional ATTOM/CoreLogic name comparison on property_data_match. ATTOM owner checks request a record at most 24 hours old. The SQL records evidence and can strengthen the claim. | Request a permitted, current, exact-unit record or use document/manual review. Keep the claimant response opaque. A similar name cannot confer ownership or occupancy. |
| **Outside this purchase** | Public Place preview uses independent public/area data. HUD rent and PMMS rate watch are separate. The deed/lien half of Record Watch is deferred. | Preserve these flows. National mortgage/lien history is a future product decision. Exact county facts and money signals keep their existing audiences. |

Two adapters need special care: the property profile currently falls back from AVM to assessor totals and derives a field named ZIP median from a single assessment. The AI profile route checks home.view. Audit that projection separately from verified-owner Property details before expanding source coverage.

Endpoints, fields and fallbacks · County record sections · Owner detail guard · Claim evidence SQL · AI route · Record Watch

# Lob moves real mail

Pantopus already generates codes and makes the access decision. Lob accepts a postcard order, produces it, enters it into the mailstream and reports what it can observe.

| Flow | Existing contract | Acceptance for another printer |
|---|---|---|
| **Canonical address verification** | Start, status, resend and confirm use an address attempt, token and MailVerificationJob. The destination includes the specific unit. A successful code can attach the eligible occupancy. | Carry the exact address and unit, operational mail type, code/verification link, non-disclosing copy, attempt limits, expiry and cooldown. Confirm against the original destination. |
| **Home verification** | Current request/confirmation/recovery and legacy Home endpoints use HomePostcardCode, admission and dispatch records. A pending owner can verify residency while ownership stays pending. | Replays must survive reloads and mobile upgrades. Retain the original provider for in-flight pieces. A receipt or tracking scan never redeems a code or approves an owner. |
| **Block Founders invite** | Verified residents can send template-only marketing postcards. The service reserves before sending, caps sends at three per sender per week, suppresses repeats for 90 days and supports a permanent recipient opt-out. | Use one durable invite/job key. The current custom-postcard call lacks verification's idempotency header and releases the reservation on an exception. Hold an ambiguous outcome for reconciliation before retrying. |
| **Status everywhere** | Signed raw-body webhooks, replay checks and vendor-event dedup route to the correct flow. Mailbox Check and the assistant also read the user's own status. Digital Mail letters/cards are separate. | Keep created/accepted, mailed/in transit, returned, rerouted, failed, unknown and code-confirmed distinct when the transport supports them. Preserve an honest coarser status when it does not. |

Qualify a pay-per-piece print/mail provider or a local printer with USPS induction and equivalent operational controls. A founder-operated small batch can bridge volume if it can meet turnaround, privacy and reconciliation requirements. PostGrid is a procurement candidate. Prove tracking, suppression and recovery on the exact service tier.

Neighbor Messages, residency letters, real-rent comparisons, verified-only Plans, selected digital Mail and block privileges depend on the resulting trust state. Verify the other participant's view even though these features make no Lob print request themselves.

Verification and custom sends · Home requests · Webhook routing · Invite reservation · Lob event semantics · PostGrid print/mail offering

# How to stand up national coverage

Give every U.S. address the same entry and recovery flow. Build one independently sourced national index and acquire the missing evidence for each kind of decision.

| Capability | Nationwide source and operating design | What must be proven |
|---|---|---|
| **Find a physical address** | Seed from the rights-cleared portions of USDOT NAD, Overture and state/local feeds. Normalize into Pantopus IDs and keep source dates and precision. Use a licensed national fallback or manual entry for missing records. | I queried the public NAD layer: 99,226,272 features. Overture lists 126,297,122 U.S. points and describes U.S. coverage as partial. Sources overlap. Neither number measures unique homes, valid units or deliverability. |
| **Check a postal destination** | Request a U.S.-wide postal processor with written app-use rights. Evaluate Google's existing integration, a licensed local engine such as Melissa, or USPS AMS under the applicable service-provider license. | Prove DPV and unit semantics, RDI/CMRA/vacancy, update cadence, field retention, territories, allowed replicas, and outage behavior. Confirm how to handle new and non-postal physical locations. |
| **Offer property facts nationally** | License a bulk national parcel/characteristics baseline, served from Pantopus where permitted. Compare Regrid for parcels/units and Cotality or First American for characteristics/transactions. State sources and verified county connectors can improve particular fields. | Obtain a county-by-field coverage file, allowed fields, sample release, SLA and explicit display/derivation/deletion rights. Price structural facts separately from ownership, tax, AVM and sales trends. Each may need a different source. |
| **Deliver a real postcard** | Operate one national fulfillment interface with a qualified primary printer and a separately qualified contingency. Pantopus owns templates, code/opt-out logic and status. | Contract for the full delivery footprint, including exceptions, return handling and cutoffs. Purchase postage per piece. Never hand an uncertain piece to a second printer until the first is reconciled. |

## Procure evidence by field and intended use

Request a downloadable coverage manifest, permitted evaluation samples, update/delta cadence, fallback rights and a written price for consumer display, local indexing, derivatives, staging/replicas and exit. Have each supplier identify upstream dependencies where possible. Two resellers of the same source cannot provide independent corroboration or outage protection.

If the policy is zero new recurring data contracts, the app can still offer nationwide entry, public facts where available, and reviewed proof. An open-only route needs ongoing source acquisition and maintenance, and must disclose gaps. Equivalent nationwide postal assurance, property depth and tracked physical mail cannot be promised on that basis.

U.S. DOT address layer · Overture U.S. count and coverage · USPS AMS · Regrid field coverage · Cotality characteristics · First American bulk delivery · Google postal response fields

# Test the nationwide promise

Start with the Census inventory of counties and equivalents. Track all 50 states and D.C.; separately exercise Puerto Rico, Guam, the U.S. Virgin Islands, American Samoa and the Northern Mariana Islands, plus APO/FPO/DPO postal routes.

| Independent evidence | National design consequence |
|---|---|
| **Florida: statewide rolls** | Florida DOR publishes real-property, sales and GIS data through a statewide program and excludes protected records. Prefer one permitted state connector where it serves multiple counties. Track each county's release and missingness. |
| **New York City: condos** | PLUTO aggregates condo units to a billing lot. A high-confidence match to that lot cannot identify apartment 4B. Require a defensible unit-level link or return parent property only. |
| **Texas: sale-price access** | Fort Bend's appraisal district explains limits on providing a list of sales. Price coverage varies by disclosure regime and source. License the field where needed or show unavailable. A tax valuation cannot fill in a missing sale. |
| **Alaska, Hawaii, rural and tribal areas** | A street line, map point, residential structure and mailbox may refer to different places. Include village, route/box and nonstandard addresses in the holdout. Obtain authorized sources for tribal jurisdictions and support confidential human review. |
| **Puerto Rico and military mail** | USPS documents urbanización as a material component in Puerto Rico. Preserve that text and postal-city aliases. Exercise APO/FPO/DPO as mailing destinations, and separate physical-property identification and proof. |
| **Washington: public-file example** | Clark's assessment summary has bedroom and bath fields; the public taxlot layer has no positive tax amount. Use field-specific joins and independent unit validation. The detailed aggregate check is in the appendix. |

## Release evidence by jurisdiction and field

For each county/equivalent record independent and fallback search coverage, postal footprint, unit evidence, parcel-link precision, each property field's filled/evidenced rate, source age, rights, and the user's recovery option. Track actual households served and the weakest jurisdiction. Publish only capability coverage; aggregate away any claim/occupancy signals.

Set a minimum sample in every state and D.C., with separate territory and rare-format cohorts. Oversample new builds, condos, rented apartments and true negatives. Distinguish an untested jurisdiction, a missing field, a stale input, an unsupported use and an actual nonmatch. Prevent a go-live claim when any required geography has only an unexamined blank.

Census county/equivalent scope · Florida DOR roll files · NYC PLUTO data dictionary · Fort Bend sales FAQ · USPS Puerto Rico urbanization · USPS military addressing

# Own the records Pantopus can defend

The combined design has four responsibilities behind the current API. They can start as modules in the existing backend and Postgres/PostGIS deployment.

| Responsibility | Own and reuse | Data boundary |
|---|---|---|
| **National address and property records** | Union approved nationwide inputs, state programs and county corrections. Use stable Pantopus IDs, historical aliases and versioned links among site, building, unit, parcel and tax account. | Restrict the index to approved fields and licenses. Retain raw snapshots only when permitted. One county record distributed via NAD and Overture counts as one source. |
| **Postal evidence** | A replaceable check for an independently supplied address. Return the checked components, unit state, usable flags, engine/version, timestamp and allowed retention. | Licensed outputs have their own purpose and expiry. A postal validation cannot populate the independent registry or prove who lives there. |
| **Private authorization** | Reuse HomeAddress, Home, HomeOccupancy, AddressClaim, ownership claims, review queues and postcard attempts. Track claim role, proof method, scope, expiry/revocation and disputes. | Household IDs, occupancy, evidence, verified unit contributions and exact destinations stay private. Search responses and public counts must not reveal who has claimed a unit. |
| **Fulfillment** | A common interface for verification postcards and custom invitations, with durable internal job identity, chosen provider, purpose, destination snapshot and event ledger. | Only the printer gets the fields needed for that piece. Only authorized users see status. Keep challenge secrets, full addresses and raw callbacks out of analytics. |

## Keep stable IDs through ordinary changes

Use source identifiers plus jurisdiction, carefully normalized address and unit, and conservative spatial joins. Exact candidates can link automatically after validation. An ambiguous condo stack, ADU or apartment resolves to a parent or review queue. Preserve old IDs through aliases and tombstones. A split requires re-resolution before any private privilege follows it.

Each fact needs its definition, units, upstream origin, jurisdiction, source observation date, effective dates, release ID, freshness and permitted audience. Store household corrections separately pending review. Preserve legacy HomeAddress UUIDs and the clients' API shapes. Any new table goes through the founder's approval and an additive migration.

Existing canonical address and aliases · Destination and unit binding · NAD coverage and use · Overture address IDs · Overture source licensing

# What people see during uncertainty

The screens can stay familiar. Most of the improvement comes from changing which result blocks a person and what the app says when it lacks evidence.

| Situation | Experience | Authorization and recovery |
|---|---|---|
| **A renter enters an unlisted apartment** | “We found the building. We couldn’t confirm apartment 4B.” Keep “4B”, allow correction or evidence, and let the person continue with permitted private setup or save their draft. | Show no occupant names, unit directory or nearby claim count. Require the existing address-proof or review path to unlock private information. |
| **The suggested street, number or unit changes** | Show entered and proposed destinations before mailing. Preserve a manually entered unit and leading zeros. Explain any material change and obtain confirmation. | Bind a code to user, unit, destination and attempt. Revalidate changed destinations. Avoid implicit transfers after a move, address split or parcel renumber. |
| **A new build, rural route, PO box, mixed-use site or protected address** | Offer location/manual review or a mailing-only address where appropriate. A business suite, shared delivery point and dwelling can each have a different role. Protected users need a confidential review path that never asks for their physical address. | Preserve residential and business rules. Withhold exact-unit privileges when evidence is insufficient. Separate public business location, private home and correspondence address. |
| **A card is delayed, returned or in doubt** | Keep the saved request across relaunch. Show the last observed state. Offer status, a permitted resend after cooldown, or existing document/landlord/admin review as applicable. | Pause automatic reissue after uncertain dispatch or rerouting. A delivery event does not identify the recipient. Wrong/expired codes and address changes must not unlock access. |
| **Only some property fields are covered** | Show dated county facts first. Let the household enter missing facts with a household label. If a value estimate is unavailable, display a labeled assessment or omit the estimate. | A pending owner can finish residency separately. Show owner records only to eligible owners. Guests, outsiders and renters retain their existing scopes. |

When a national source has a local gap, keep the same address form, preserve input and retry with a licensed fallback or review. The separate business geocoder and suite heuristics also need cross-regression. Measure abandonment, unit recovery, time to a useful Place screen, successful proof and review age by geography.

Instant claim access boundary · Postal recovery on iOS · Mailbox Check · Washington protected substitute addresses

# Separate the fast path from the proof

Search and ordinary Place reads should succeed when postal systems, property suppliers or the printer are down. Availability means a usable answer with the correct permissions.

| Service objective | Proposed U.S. launch target | How to demonstrate it |
|---|---|---|
| **Address search** | Nearest healthy serving region: warm server p95 ≤150 ms and p99 ≤400 ms. Target device p95 ≤350 ms on the named test network, excluding debounce. | Measure coasts, interior, Alaska, Hawaii and territories separately on web and real mobile clients. Test cold/warm at twice projected national peak. Report empty results and abandonment. |
| **Selected address / property** | Local resolve p95 ≤300 ms. Authorized cached property read p95 ≤400 ms. Render known fields immediately. Expensive enrichment continues separately. | Compare covered and uncovered areas, multifamily and rural cohorts. Cache by release, jurisdiction and permitted audience; never share a private response in a public cache. |
| **User-facing availability** | Proposed 99.9% over 30 days for eligible search and read requests, about 43 minutes of error budget. Track complete answers separately from partial/unknown ones. | Count outages and excess unknowns per feature. Add independent health probes, replica/failover and restore drills. A single shared host cannot substantiate this target on its own. |
| **Postal and mail** | Give the UI a durable pending state when work is queued. Bound external attempts by deadline. Measure provider acceptance, dispatch lag, delivery/return and code redemption independently. | Circuit-break failing dependencies, back off with jitter and reconcile uncertain sends before failover. Alert on queue age, returns, late callbacks, duplicate pieces and duplicate grants. |
| **Recovery and freshness** | Keep the previous permitted release ready; rehearse restoring search within 30 minutes. Acknowledged proof/opt-out writes require durable storage and an idempotent key. | Monitor per-source release dates nationwide. Rehearse lost worker/region, bad import, expired postal data and database restore. Reapply erasures before serving restores. |

Serve immutable search releases from at least two independently failover-tested U.S. locations, within license limits. Partition by jurisdiction with cross-border search. Replicate the authorized private store across failure zones and fail closed if a proof write cannot be committed. Roll back by release. A stale permitted public fact can carry a date; expired postal/ownership evidence cannot authorize access.

Existing validation metrics · External cache behavior · USPS maintenance lifecycle · Lob webhook delivery

# Resolve usage rights before moving data

There is an immediate retention question, separate from the long-term decision to replace the vendors. The applicable Pantopus order forms and any negotiated terms still need to be established.

## ATTOM has several storage paths

ATTOM's published evaluation terms restrict use to evaluation, prohibit building databases from the results, cap retention at 24 hours and specify purge/certification at termination. A previous November 1–2 prefill recorded 195 saved trial responses from 200 attempts. That is a historical result. The current contents of hosted systems and any overriding agreement were not checked in this review.

In the audited code the request cache writes an infinite expiry and property/detail can be read with no age limit. Home.niche_data can hold a complete property-detail snapshot. AttomPropertyCache, PropertyIntelligenceCache, PlaceSectionCache and stale fallback reads add copies or derivatives. Smarty defaults to a 90-day candidate cache in HomeAddress and also persists derived flags, provider versions and raw material.

First founder action: suspend further trial prefill and avoid using those outputs for the new registry or evaluation. Inventory raw/derived stores, secondary indexes, exports, logs, backups and offline reports. Verify the signed rights, then arrange any required quarantine, deletion or certification. Keep an independent household entry and associated access evidence distinguishable from restricted vendor data.

## Postal, county and open-data rights

Smarty's published §§6–7 cover duration, competitive development, benchmarking and compiling databases. Enrichment has additional consumer/household restrictions. Check the exact subscribed product and order overrides, especially before enabling the optional secondary lookup. Review Lob's competitive-use restrictions. Build and evaluate only with independently permitted data and adjudication.

USPS AMS is a locally operated licensed product with periodic reference/software updates. CASS certifies address-matching software. Direct DPV licensing has manufacturer eligibility. The hosted USPS product has separate terms. Confirm field of use, display, retention, replicas, upgrades and exit terms in writing. Google or an on-premises engine may bridge the product only if coverage and rights fit.

Limit the county import to approved non-personal fields. Washington restricts commercial requests for lists of individuals. Metro's open-data license and Overture's per-source terms require their own review. Keep resident proof, owners' names, tax payment details and protected addresses out of the public index. Make the release reproducible using cleared sources alone.

ATTOM use and termination terms · Smarty agreement · Smarty product terms · Lob terms · USPS AMS license · Hosted USPS license · Washington statute · Persistent property reads

# Move one capability at a time

Bring up the national fallback and independent coverage matrix before broad cutover. Roll out by capability on the nationwide service; canaries control risk within that rollout.

| Stage | Work and decision gate |
|---|---|
| **Week 1 · establish rights and baseline** | Confirm order forms, current live configuration and current bills. Count distinct addresses, Google/Smarty checks and secondary calls, ATTOM endpoint requests/cache hits, mail pieces/retries/returns, user completions and review hours. Inventory every old client and stored copy. |
| **Weeks 2–3 · prove national inputs** | After clearance, stage national address and property releases. Obtain samples and rights for missing jurisdictions and the precise property fields. Exercise urban/rural, multifamily, tribal, military and territory cases across the country. Keep enrichment and enforcement independently gated. |
| **Week 4 · adjudicate the gaps** | Use 500–1,000 adjudicated cases to identify failure modes. Allocate a floor to each state/DC and separately to territories and rare formats. Expand until the uncertainty bounds meet the production gates. Hold out buildings, jurisdictions and source releases. Use independent evidence. |
| **ATTOM exit gate** | Required national fields meet per-geography coverage and match thresholds or have a contractual backstop. The founder accepts any residual gaps. Replace key-based Place/exemption checks, ATTOM labels, AI outputs, SQL provider allowlists and native readers. Retire stored copies lawfully. |
| **Smarty exit gate** | Replace required postal health checks in validate, unit revalidate, Home create, claim reconstruction and Mailbox Check. Exercise D/N1, S/unknown, blanks, outage, PO box, CMRA, vacancy, mixed use, genuine missing units, conflicts and cached/mismatched addresses. Verify unit/role boundaries. |
| **Lob exit gate** | A founder-controlled, consented trial proves end-to-end printing and code redemption in representative regions. Confirm coverage for remote geographies in writing; exercise delays, returns, unknowns, duplicates and opt-out. Drain old jobs and callbacks without double-mailing. |

Canary by jurisdiction and capability, with independent rollback switches. Preserve legacy UUIDs and stored claims across cutovers. iOS and Android already allowlist specific source values; deploy compatible readers first. Never label a new county source “ATTOM” to placate older builds. Withhold unsupported fields for old clients or negotiate a versioned response.

Every jurisdiction needs a route to submit, correct and seek review. Proposed automated-coverage floor: resolve ≥95% of independently confirmed residential premises to the building without staff in that jurisdiction. Publish misses and the share needing manual help; an unknown result is an abstention.

iOS source allowlist · Android source allowlist · Place credential gate · Exemption gate · Licensed local processor example

# What would justify canceling

A national rollout earns its claim through the worst-served groups as well as the national total. Cost savings only count if users can still complete the intended job.

## Quality and real-app acceptance

Proposed targets: ≥98% top-five search recall and ≥99.5% precision for automatic property links on the national holdout. Report each state, territory, address type and urban/rural split, plus population-weighted and worst-group results. Report abstention and coverage separately. Target <0.1% false acceptance of adjudicated invalid units. With zero errors, about 3,000 independent invalid cases are needed for a one-sided 95% upper bound near 0.1%. Repeated variants of one building do not supply that evidence.

On web, iOS and Android with a real local backend, exercise add/reopen, correct the unit, claim/review, request/resend/redeem, move/revoke, owner and renter views, neighbor invite and opt-out. Repeat across relaunch and reload, losing network, out-of-order events and an old client. Inspect the other household's view. The founder alone should run live mail to consenting recipients or authorize vendor/hosted changes.

## Price the outcomes separately

Use actual monthly invoices and billable events. Separate fixed minimums and overages for address checking, property detail/AVM/trend, and printing/postage. Include legal rights, data refresh, engineering, GIS review, support, storage, replicas and fallback calls. Track cost per successful Add Home, verified unit, fulfilled card and useful property record. Register which features the cheaper option drops.

Break-even monthly volume equals incremental monthly fixed cost divided by net saving per genuinely displaced billable event. If that saving is zero or negative there is no volume break-even. Compare nationwide owned operation against the cheapest rights-compatible national alternative and the current stack. The Word model's MAU assumptions assign spend to APIs Pantopus does not call; the PDF's 2–4 week feasibility and 8–12 week beta are planning estimates. Those regional estimates do not size national acquisition, ingestion, licensing or operations. Use observed national volumes, support capacity and quotes.

> Decision now: fund national source diligence and a nationwide address-to-Home prototype. Seek a field-level national property license, qualify a postal processor and a printer in parallel, and keep an honest recovery path in every jurisdiction. End each incumbent only when its national replacement passes. Clark is one supplier test.

Review basis: master 1c5ddb658c, two supplied proposals, primary sources, synthetic parser/decision checks, a nationwide address-layer count and aggregate Clark checks. Production entitlements, live mail, cross-device journeys, benchmarks and the replacement itself were outside this read-only assessment.

Audited Git revision · Vendor prefill importer · Existing Home mail flows · Google use policy · USPS pricing/licensing categories

# One county’s source files

Clark County, Washington, is a source-validation example. These measurements cannot be extrapolated to the nation. Assessment IDs, taxlot features and households each count different things.

## Assessor building summary

I read the November 3, 2026 PACS release README and aggregated its building-summary file. It contains one row per distinct prop_id in this file: 245,518 rows and 245,518 IDs. Positive values establish field availability in this extract. They do not establish correctness or dwelling-level coverage.

| PACS field | Positive values | Meaning for the app |
|---|---|---|
| **Bedrooms** | 171,886 / 245,518 (70.0%) | 73,632 rows have zero. Preserve zero as a source value until its meaning for that property type is known. |
| **Full baths** | 171,968 / 245,518 (70.0%) | Half baths and three-quarter baths have separate columns. Display components or a documented conversion. |
| **Living area** | 178,623 / 245,518 (72.8%) | A better candidate for living area than the taxlot layer’s generic building square footage. Confirm the join and definition. |
| **Year built** | 178,744 / 245,518 (72.8%) | Keep original and effective construction years distinct. Show the dated source where a value exists. |

## Public taxlot layer

Official count-only queries returned 196,535 features. Among the 172,378 with a positive building count, 171,975 had a positive year built (99.77%). Only 10,665 of all features had a nonempty SitusUnit. That is 5.43% of features, a poor proxy for an apartment inventory. No feature had a positive TaxAmount. The separate tax-history source is necessary for tax bills.

The separate PACS tax-history header contains property ID, statement ID, valuation/display years and amount billed. The sales file has price, type and an invalid-sale flag. Next, verify year selection, valid arm’s-length sales, one-to-many joins and protected fields before any member-facing use. These checks did not measure address-to-unit match accuracy.

The adjacent Portland Metro source has different joins: taxlots marked HAS_MANY need companion records and the open product excludes ownership details. These examples show why the nationwide ingestion contract needs a per-source join specification and column-level rights.

Clark assessor and refresh policy · PACS source archive · Public taxlot schema and query service · Metro RLIS · Metro multiple-record guidance · Metro open-data license